

Norway

SLEIPNER

Volve F

F-15

F-15C

Design: F-15C

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Site:	Volve F	MD Reference:	Inspirer Rotary Table @ 54.90m (Actual RTE)
Well:	F-15	North Reference:	Grid
Wellbore:	F-15C	Survey Calculation Method:	Minimum Curvature
Design:	F-15C	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-15					
Well Position	+N/-S	-3.17 m	Northing:	6,478,560.36 m	Latitude:	58° 26' 29.7063 N
	+E/-W	3.53 m	Easting:	435,053.55 m	Longitude:	1° 53' 15.1494 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-15C				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2008	31.12.2008	-2.06	71.62	50,396

Design	F-15C				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	2,544.30
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.17	3.53	200.34	

Survey Program	Date	03.03.2016			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
153.40	1,364.50	12 1/4" F-15 Drop Gyro (F-15)	Wellbore Surveyor, stat	Gyro Tool from GD	
1,383.00	2,513.00	12 1/4" Drop Gyro F-15A Revised (F-15A)	Wellbore Surveyor, stat	Gyro Tool from GD	
2,563.48	2,803.11	12 1/4" Drop Gyro F15C (F-15C)	Wellbore Surveyor, stat	Gyro Tool from GD	
2,804.00	2,892.40	12 1/4" MWD F15C (F-15C)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
2,939.74	3,211.78	8 1/2" MWD F15C (F-15C)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.17	3.53	0.00	0.000	0.000	0.000	
153.40	0.15	253.69	153.40	-3.17	3.52	0.01	0.600	0.600	0.000	
234.20	0.28	146.02	234.20	-3.37	3.53	0.18	0.132	0.048	-39.976	
274.60	0.78	136.21	274.60	-3.65	3.78	0.36	0.376	0.371	-7.285	
315.00	1.46	128.29	314.99	-4.16	4.37	0.64	0.517	0.505	-5.881	
355.30	3.41	124.07	355.25	-5.15	5.77	1.08	1.457	1.452	-3.141	
395.70	5.13	126.98	395.54	-6.91	8.21	1.89	1.287	1.277	2.161	

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Wellbore:	F-15C	Survey Calculation Method:	Minimum Curvature
Design:	F-15C	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
436.10	5.52	139.07	435.76	-9.47	10.92	3.34	0.880	0.290	8.978	
474.40	5.62	152.33	473.89	-12.52	13.00	5.48	1.009	0.078	10.386	
514.80	6.19	163.05	514.07	-16.36	14.55	8.53	0.920	0.423	7.960	
557.10	7.03	171.79	556.09	-21.10	15.59	12.62	0.928	0.596	6.199	
597.50	7.74	174.05	596.16	-26.25	16.22	17.23	0.570	0.527	1.678	
637.80	8.49	174.54	636.05	-31.91	16.79	22.34	0.561	0.558	0.365	
678.20	8.94	175.57	675.99	-38.01	17.31	27.88	0.354	0.334	0.765	
718.50	9.34	177.45	715.77	-44.40	17.70	33.74	0.372	0.298	1.400	
758.90	10.22	185.51	755.59	-51.24	17.50	40.22	1.207	0.653	5.985	
799.30	11.89	193.59	795.24	-58.86	16.18	47.82	1.689	1.240	6.000	
839.70	13.68	199.13	834.64	-67.42	13.64	56.73	1.610	1.329	4.114	
880.00	15.14	201.54	873.67	-76.81	10.14	66.76	1.175	1.087	1.794	
920.40	13.72	203.30	912.79	-86.12	6.31	76.82	1.103	-1.054	1.307	
960.80	14.53	201.31	951.97	-95.24	2.57	86.67	0.701	0.601	-1.478	
1,001.10	16.51	200.62	990.80	-105.31	-1.28	97.45	1.480	1.474	-0.514	
1,041.60	16.33	201.19	1,029.65	-116.01	-5.37	108.90	0.179	-0.133	0.422	
1,081.90	16.18	205.25	1,068.34	-126.37	-9.81	120.16	0.853	-0.112	3.022	
1,122.30	17.54	211.88	1,107.01	-136.63	-15.43	131.73	1.747	1.010	4.923	
1,162.60	17.66	221.21	1,145.43	-146.39	-22.66	143.40	2.100	0.089	6.945	
1,203.00	16.05	230.28	1,184.10	-154.57	-31.00	153.96	2.286	-1.196	6.735	
1,243.30	15.33	232.77	1,222.90	-161.35	-39.52	163.29	0.734	-0.536	1.854	
1,283.80	18.17	237.03	1,261.68	-168.03	-49.09	172.87	2.291	2.104	3.156	
1,324.10	21.21	241.08	1,299.62	-174.98	-60.74	183.44	2.479	2.263	3.015	
1,364.50	24.71	243.03	1,336.81	-182.34	-74.67	195.19	2.659	2.599	1.448	
1,383.00	26.95	242.95	1,353.46	-186.01	-81.85	201.11	3.633	3.632	-0.130	
1,406.00	25.71	234.70	1,374.08	-191.26	-90.57	209.07	5.035	-1.617	-10.761	
1,445.00	24.44	222.89	1,409.43	-202.07	-102.97	223.51	3.966	-0.977	-9.085	
1,487.00	21.59	210.72	1,448.10	-215.09	-112.84	239.15	3.952	-2.036	-8.693	
1,544.00	20.56	192.78	1,501.35	-233.88	-120.42	259.41	3.425	-0.542	-9.442	
1,584.00	21.08	181.10	1,538.75	-247.93	-122.11	273.17	3.133	0.390	-8.760	
1,625.00	21.07	170.97	1,577.02	-262.59	-121.10	286.56	2.662	-0.007	-7.412	
1,665.00	20.92	159.85	1,614.38	-276.40	-117.51	298.26	2.986	-0.112	-8.340	
1,705.00	22.16	151.36	1,651.59	-289.73	-111.43	308.65	2.513	0.930	-6.367	
1,746.00	24.79	147.70	1,689.20	-303.78	-103.13	318.94	2.199	1.924	-2.678	
1,786.00	27.24	144.45	1,725.15	-318.32	-93.32	329.17	2.125	1.837	-2.437	
1,827.00	27.70	143.46	1,761.52	-333.61	-82.19	339.63	0.474	0.337	-0.724	
1,867.00	28.14	143.47	1,796.87	-348.66	-71.04	349.87	0.330	0.330	0.007	
1,907.00	27.08	143.10	1,832.31	-363.52	-59.96	359.95	0.805	-0.795	-0.277	
1,948.00	27.31	143.53	1,868.78	-378.55	-48.77	370.15	0.221	0.168	0.315	
1,988.00	27.71	143.72	1,904.26	-393.42	-37.81	380.29	0.307	0.300	0.142	
2,028.00	27.86	145.00	1,939.64	-408.58	-26.95	390.72	0.461	0.112	0.960	
2,069.00	27.42	149.82	1,975.97	-424.59	-16.70	402.18	1.667	-0.322	3.527	
2,109.00	26.74	154.96	2,011.59	-440.70	-8.26	414.35	1.827	-0.510	3.855	

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Survey										
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2,149.00	25.09	162.89	2,047.58	-456.97	-1.96	427.41	2.876	-1.237	5.947	
2,190.00	24.05	170.43	2,084.87	-473.52	1.99	441.56	2.415	-0.761	5.517	
2,230.00	24.40	174.59	2,121.35	-489.78	4.13	456.07	1.307	0.262	3.120	
2,270.00	23.87	181.76	2,157.87	-506.10	4.66	471.18	2.233	-0.397	5.377	
2,311.00	24.72	188.97	2,195.24	-522.86	3.06	487.46	2.256	0.622	5.276	
2,351.00	25.05	194.20	2,231.53	-539.34	-0.32	504.08	1.669	0.247	3.922	
2,392.00	25.95	202.77	2,268.55	-556.03	-5.92	521.68	2.776	0.659	6.271	
2,432.00	27.22	209.99	2,304.33	-572.03	-13.88	539.44	2.602	0.952	5.415	
2,486.00	28.73	219.32	2,352.04	-592.78	-28.29	563.91	2.569	0.839	5.183	
2,513.00	30.35	223.67	2,375.54	-602.73	-37.11	576.31	2.985	1.800	4.833	
2,563.48	33.26	229.77	2,418.44	-620.91	-56.50	600.09	2.575	1.729	3.625	
2,603.80	32.08	231.01	2,452.38	-634.78	-73.26	618.93	1.009	-0.878	0.923	
2,644.15	27.57	233.84	2,487.38	-647.04	-89.14	635.94	3.512	-3.353	2.104	
2,684.45	25.99	233.83	2,523.36	-657.76	-103.79	651.08	1.176	-1.176	-0.007	
2,724.77	22.08	234.57	2,560.17	-667.37	-117.11	664.72	2.918	-2.909	0.551	
2,765.09	16.67	243.57	2,598.21	-674.34	-128.47	675.21	4.587	-4.025	6.696	
2,803.11	13.75	248.58	2,634.89	-678.42	-137.56	682.19	2.525	-2.304	3.953	
2,852.00	11.69	248.05	2,682.58	-682.39	-147.57	689.39	1.266	-1.264	-0.325	
2,892.40	12.47	247.23	2,722.08	-685.61	-155.38	695.13	0.593	0.579	-0.609	
2,939.74	12.77	250.95	2,768.28	-689.30	-165.04	701.94	0.549	0.190	2.357	
2,969.76	13.62	261.50	2,797.51	-690.90	-171.68	705.75	2.548	0.849	10.543	
2,997.14	15.39	260.20	2,824.02	-692.00	-178.45	709.13	1.972	1.939	-1.424	
3,038.44	18.98	261.37	2,863.47	-693.94	-190.49	715.14	2.620	2.608	0.850	
3,078.27	19.82	259.28	2,901.04	-696.17	-203.53	721.76	0.821	0.633	-1.574	
3,118.28	21.94	256.07	2,938.42	-699.23	-217.45	729.47	1.806	1.590	-2.407	
3,158.54	22.35	256.20	2,975.71	-702.87	-232.18	738.00	0.308	0.306	0.097	
3,199.02	21.27	256.14	3,013.29	-706.46	-246.79	746.45	0.801	-0.800	-0.044	
3,211.78	20.98	255.56	3,025.19	-707.58	-251.25	749.05	0.840	-0.682	-1.364	
3,232.00	20.98	255.56	3,044.07	-709.39	-258.26	753.18	0.000	0.000	0.000	

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name			Casing Diameter (in)	Hole Diameter (in)
220.60	220.60	30"			30.000	36.000
1,368.40	1,340.35	20"			20.000	26.000
2,561.80	2,417.04	13 3/8"			13.375	16.000
2,914.80	2,743.95	9 5/8"			9.625	12.250
3,231.00	3,043.14	7"			7.000	8.500

Checked By: _____ Approved By: _____ Date: _____